

Infosys Springboard Virtual Internship 6.0

Completion Report

Name: Kishore Kumar H

Batch Number 4

Start date 13th October 2025

Internship Duration: 8 Weeks

1. Project Title

AI-Based Image Transformation Tool for Cartoon Effect Generation.

2. Project Objective

The objective of the project is to design and implement an **AI-driven image transformation platform** that enables users to upload an image and convert it into a selected artistic style such as **Pencil sketch, Oil Painting, Pixar, Studio Ghibli, Comics, and 2D Cartoon**. The system uses Machine Learning models, including **Stable Diffusion** and **ONNX-based style models**, to generate high-quality outputs while ensuring secure user authentication, optimized processing performance, and seamless cloud integration.

3. Project description in detail

During my internship, I developed an **AI-driven image transformation platform** that allows users to upload an image and convert it into a selected artistic style such as **Pencil Sketch, Oil Painting, Pixar, Studio Ghibli, Comics, and 2D Cartoon**. The system intelligently combines **Machine Learning-based models** with **traditional computer vision techniques** to deliver high-quality stylized outputs depending on the selected style.

For styles such as **Pixar, Comics, and 2D Cartoon**, the platform uses **Stable Diffusion-based ML models**, while the **Studio Ghibli** style is generated using an **ONNX-based model**. Lightweight transformations like **Pencil Sketch** and **Oil Painting** are implemented using **OpenCV**, ensuring fast execution without heavy ML inference.

The platform includes a **React-based frontend** for image uploads, style selection, and displaying results, along with a **Flask backend** that manages model execution, authentication, payments, and data processing. Users can **instantly download generated images** and later access them through a dedicated **History section**, where they can also delete unwanted outputs.

A **credit-based system** is implemented for learning purposes: each new user receives **5 free generations**, and a **dummy payment module** simulates credit purchase through mock **Card** or **UPI** options. Upon successful simulated payment, **5 additional credits** are added to the user account. This module was created to understand the integration and workflow of payment features within web applications.

The system integrates **SQLite Cloud** for remote database management and supports **secure user authentication** and **optimized processing**, ensuring stability, performance, and reliable cloud connectivity.

Approach:

1. **Requirement Analysis & System Planning:** Identified the need for a hybrid image transformation system combining **ML** and **computer vision**. Defined core modules such as Model Processing, Authentication, History Management, Credit System, Frontend UI, and Cloud Storage. Planned a balanced architecture to ensure quality and performance.
2. **Model Evaluation & Selection:** Evaluated different image-to-image models and finalized **Stable Diffusion** for high-quality styles like Pixar, Comics, and 2D Cartoon. Chose an **ONNX-based model** for Studio Ghibli due to faster inference. Used **OpenCV** for Pencil Sketch and Oil Painting to reduce computation time. This ensured the best model choice for each style.
3. **Backend Development (Flask):** Designed **REST APIs** for authentication, image uploads, model inference, credit updates, and history retrieval. Implemented separate processing functions for Stable Diffusion, ONNX, and OpenCV styles. Ensured backend security using **bcrypt** and proper API validation.
4. **Frontend Development (React):** Built an **interactive UI** for uploading images, selecting styles, viewing results, accessing history, and managing credits and dummy payments. Established smooth communication with the backend using REST APIs.
5. **User Authentication & SQLite Cloud Integration:** Implemented session-based authentication logic with secure **password hashing**. Connected to **SQLite Cloud** to store user accounts, credits, and history logs, ensuring persistent cloud-based access.
6. **Credit System & Payment Simulation:** Developed a **credit deduction mechanism** where each transformation consumes 1 credit. Created a **dummy payment module** to simulate credit purchases via mock Card/UPI flows. Automatically added **5 credits** after each simulated payment.
7. **History Module Implementation:** Stored and retrieved generated images from the cloud database. Enabled users to instantly download outputs after generation and manage **(view/delete)** their complete history.
8. **Performance Optimization:** Used **lazy model loading** to speed up startup time. Applied **in-memory image handling** and **basic caching** to reduce processing delays, optimized ML inference and OpenCV operations for smoother output generation.
9. **Testing & Iterative Refinement:** Tested different image formats, resolutions, and styles to ensure compatibility. Fixed issues related to processing delays, credit updates, and history retrieval. Refined both frontend and backend for consistent performance.

Technology Stack:

1. Frontend:
 - I. **React.js** – For building the user interface and handling routing and component-based structure.
 - II. **Tailwind CSS** – For fast, utility-based styling and responsive UI design.
 - III. **JavaScript (ES6+)** – Core scripting for frontend logic.
 - IV. **HTML5** – For structuring the UI components in React.
2. Backend:
 - I. **Flask (Python)** – REST API development for model processing, authentication, credit management, and history handling
 - II. **Bcrypt** – For secure password hashing
3. Machine Learning & Image Processing Models:
 - I. Stable Diffusion – **Nitrosocke/Classic-Anim-Diffusion** – Used for Pixar and 2D Cartoon Style transformations.
 - II. Stable Diffusion – **Dreamshaper8** – Used for Comic Style output with high detail and artistic tone.
 - III. **AnimeGAN (ONNX Model)** – Used for Studio Ghibli style due to lightweight inference and faster processing.
 - IV. **OpenCV** (inversion + Gaussian blur technique) – Used for Pencil Sketch
 - V. **OpenCV** (xphoto module) – Used for Oil Painting
4. Database:
 - I. **SQLite Cloud** – Cloud-hosted database for remote access
5. Tools & Support:
 - I. **Git / GitHub** – For version control and project source code storage.
 - II. **ngrok** – Used to expose both backend and frontend when running the project on Google Colab
 - III. **Google Colab** – Used to run the Flask backend, ML models, and GPU-accelerated image transformation
 - IV. **Cursor / Kiro** – Used as an AI-powered coding assistant for improved developer productivity.
 - V. **ChatGPT / Claude** – Used for debugging assistance, optimization guidance and reviewing implementation logic.

Real-World Impact:

This project demonstrates how AI-based image transformation can be applied in real-world creative and digital industries. Platforms using similar technology enable users to generate stylized artwork for **design, gaming, animation, social media content, and marketing** without requiring advanced artistic skills. The hybrid approach of **ML models** and **OpenCV** techniques ensures both **high-quality outputs** and **fast processing**, making the system suitable for integration into **mobile apps, web editors, and creative tools**. The **credit system, history management, and payment simulation** also reflect how modern AI services operate commercially, providing a realistic model for **scalable AI-driven products**.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Implement Pencil Sketch Style using OpenCV, Pillow, and SciPy.	Successfully completed Pencil Sketch transformation using OpenCV along with Pillow and SciPy.
Week 2	Oil Painting style using OpenCV stylization, xphoto, and Stable Diffusion (runway-ml-5).	Implemented Oil Painting style using OpenCV stylization, xphoto, and Stable Diffusion; tested and finalized the output.
Week 3	Implement Pixar and 2D Cartoon style transformations using Stable Diffusion models.	Successfully implemented Pixar and 2D Cartoon styles using Nitrosocke/Classic-Anim-Diffusion.
Week 4	Implement Comics and Ghibli styles using Stable Diffusion and AnimeGAN.	Comics and Anime style generations are completed and the outputs are finalized.
Week 5	Frontend UI creation and backend API integration.	Built React-based UI and fully integrated with Flask backend.
Week 6	Implement History and Payment Features, run full project on Google Colab and expose services using ngrok.	Added history module and dummy payment system, successfully executed the entire project on Google Colab, and exposing both the frontend and backend using ngrok.
Week 7	Mock presentations and final polishing of project flow.	Conducted mock presentations and refined the project for final submission.
Week 8	Complete project documentation and prepare for final presentation.	Completed full project documentation and prepared for the final presentation.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Introduction to GitHub and ChatGPT usage, along with an overview of Milestone 1.	14 th Oct 2025
Prototype/First Draft	Implemented Pencil Sketch and Oil Painting styles using OpenCV and image processing techniques.	27 th Oct 2025
Mid-Term Review	Developed and refined Pixar, 2D Cartoon, Anime and Comics styles using ML models and finalized the outputs.	10 th Nov 2025
Final Submission	Completed the UI & backend, ran the project on Google Colab, and exposed it via ngrok.	24 th Nov 2025
Presentation	Presentation / Demonstration	Awaiting confirmation.

5b. Project execution details

The entire project was executed on **Google Colab**, enabling **GPU-accelerated model inference** and seamless integration of both **frontend and backend** services. The execution process consisted of four major steps.

1. Environment Setup & Repository Initialization:

- I. Installing **Git**, **Node.js**, and **npm** for running the React frontend.
- II. Cloning the complete project repository from **GitHub**.
- III. Installing backend dependencies such as **Flask**, **Flask-CORS**, **bcrypt**, **SQLite Cloud SDK**, **ONNX Runtime**, and **accelerate**.
- IV. Verifying **Node**, **npm**, **CUDA** availability, and ensuring all system-level packages are correctly configured.

2. Node.js Reinstallation for Frontend Compatibility:

- I. Removed existing Node.js/npm versions to avoid conflicts with Vite.
- II. Installing **Node.js v18 LTS**, which is required for stable Vite + React development.

3. Backend Setup, Launch & ngrok Exposure:

- I. Configured **ngrok authentication** for secure tunneling.
- II. Started the **Flask backend** silently on port 3001.
- III. Exposed the backend publicly using **ngrok**, generating a live URL accessible from anywhere.
- IV. Saved the backend URL into a file so the frontend could automatically update and communicate with it.
- V. Ensured **GPU-enabled backend inference** for Stable Diffusion, ONNX, and OpenCV pipelines.

4. Frontend Build, Launch & Public Exposure:

- I. Dynamically updated the **BACKEND_URL** inside the Dashboard.jsx using the ngrok backend URL.
- II. Started the **Vite-powered React frontend** silently on port 5173.
- III. Exposed the frontend via a second **ngrok tunnel**, making the full application (UI + Backend) accessible over the internet.
- IV. Ensured both **backend and frontend** ran simultaneously inside Google Colab.

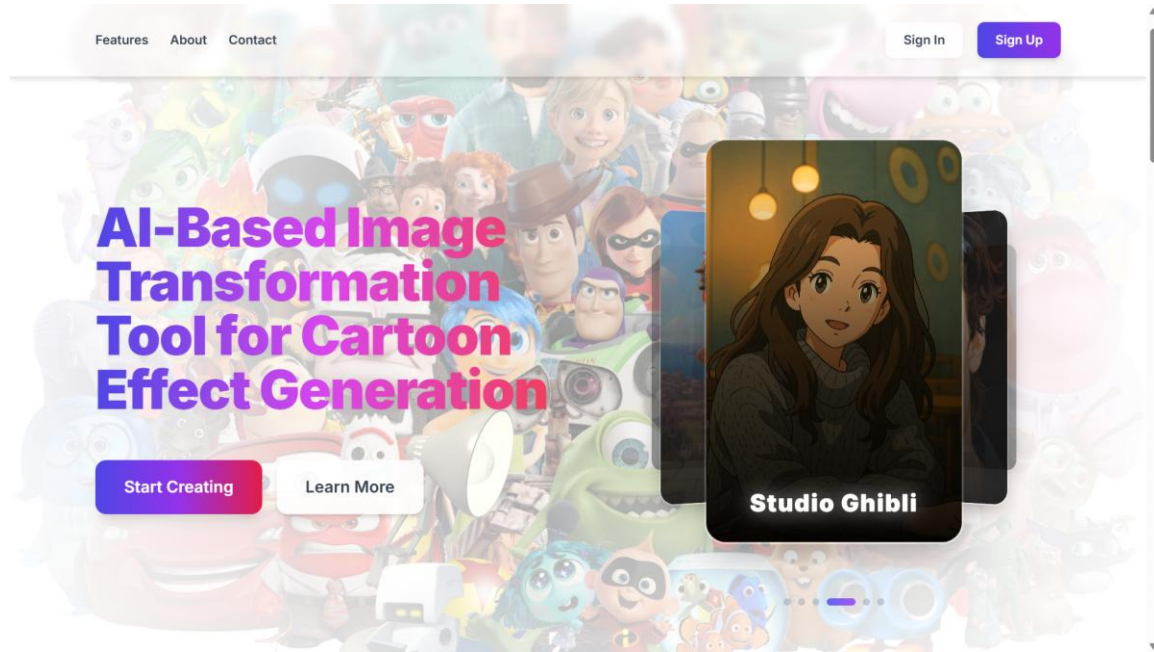
5. Model Configuration, Monitoring & Debugging:

- I. Configured essential **inference parameters**:
 - a. **Strength**: 0.4 – 0.6
 - b. **Guidance Scale**: 7.5 – 8.0
 - c. **Input Resolutions**: 728×728 (Stable Diffusion), 512×512 (AnimeGAN)
- II. Included dedicated cells to view **backend logs (backend.log)** and **frontend logs (frontend.log)**.
- III. Monitored backend logs to verify **model loading** (Stable Diffusion, ONNX, OpenCV), **API requests**, **history saving**, and **credit updates**.

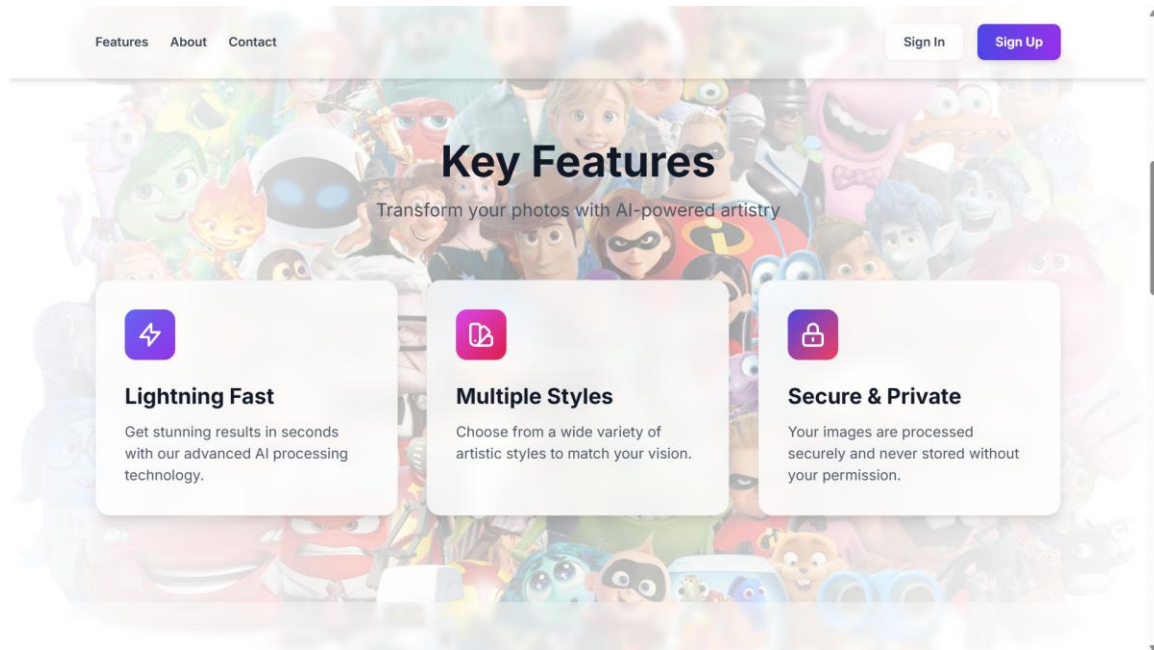
6. Snapshots / Screenshots

Landing Page

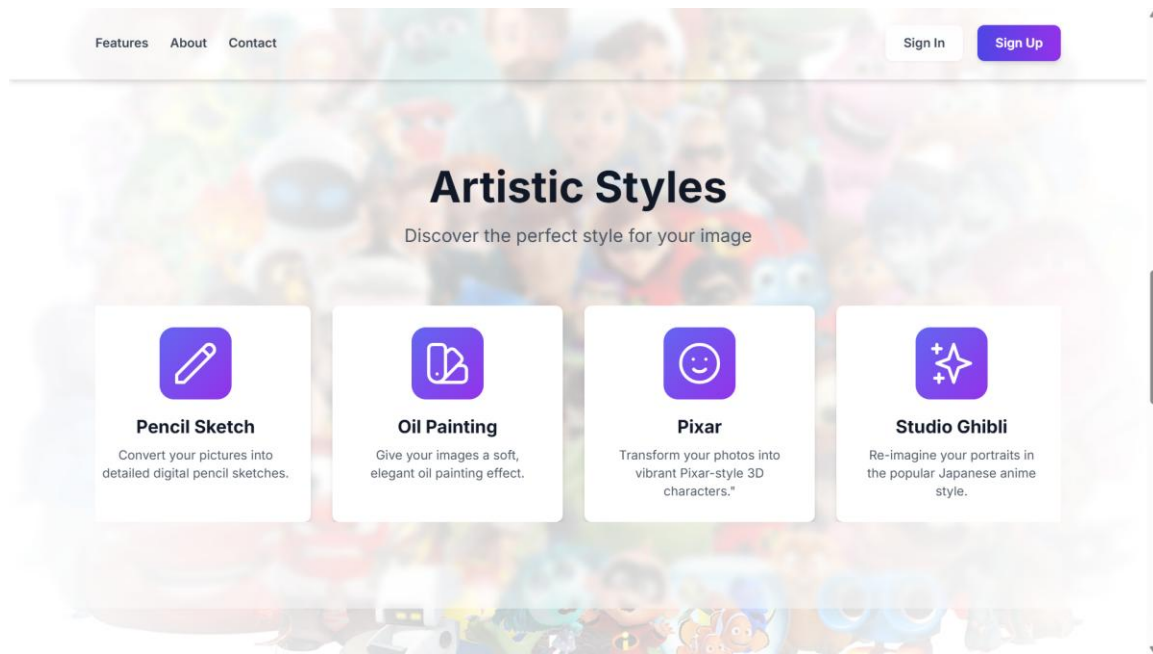
Hero Section



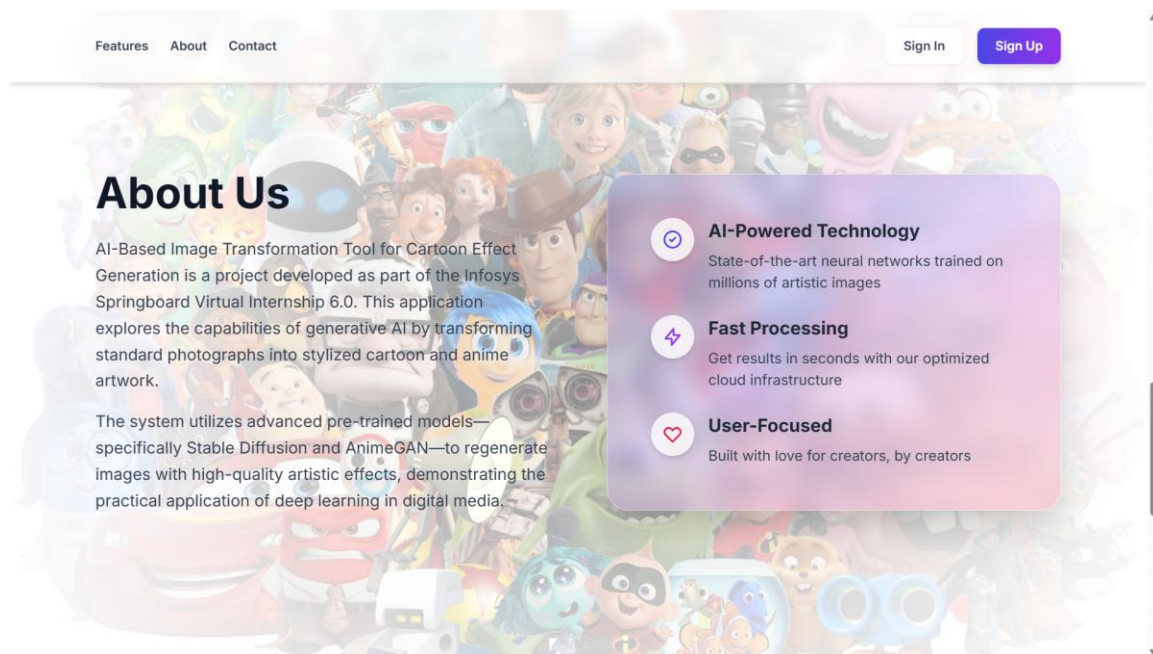
Features Section



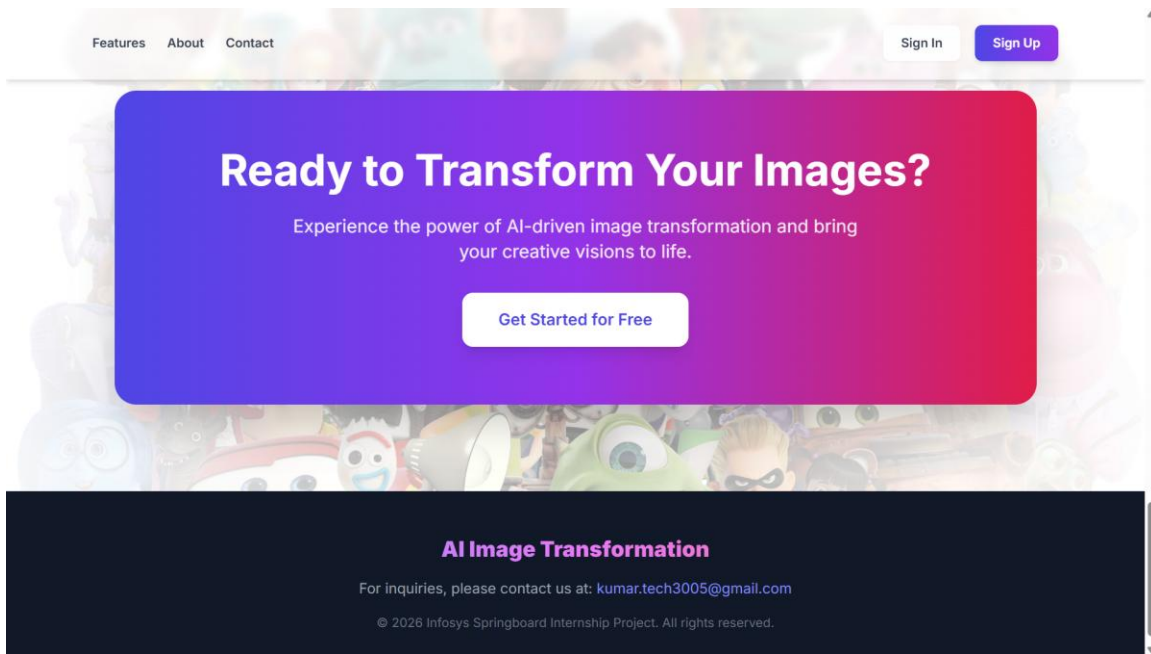
Artistic Styles Section



About Us Section



Contact Section



This mockup shows the contact section of a website. At the top, there is a navigation bar with links for 'Features', 'About', and 'Contact', along with 'Sign In' and 'Sign Up' buttons. The main content area features a large, colorful gradient box with the heading 'Ready to Transform Your Images?' and a subtext 'Experience the power of AI-driven image transformation and bring your creative visions to life.' Below this is a 'Get Started for Free' button. The background of the page is a collage of various cartoon and movie characters. At the bottom, there is a dark blue footer with the text 'AI Image Transformation', contact information 'For inquiries, please contact us at: kumar.tech3005@gmail.com', and a copyright notice '© 2026 Infosys Springboard Internship Project. All rights reserved.'

Features About Contact Sign In Sign Up

Ready to Transform Your Images?

Experience the power of AI-driven image transformation and bring your creative visions to life.

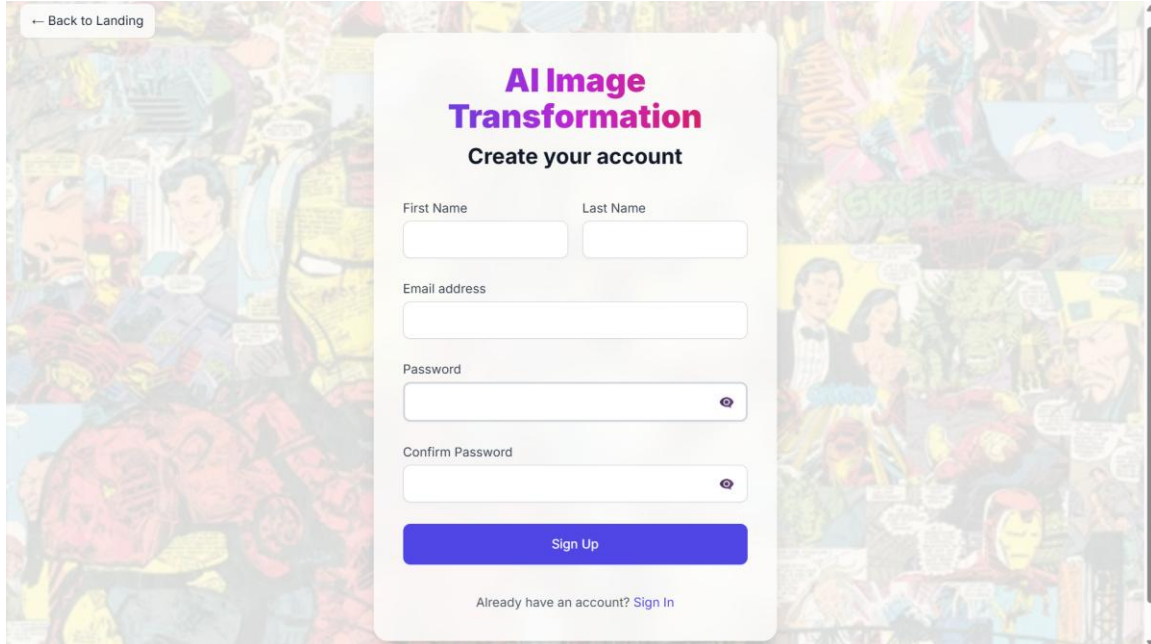
Get Started for Free

AI Image Transformation

For inquiries, please contact us at: kumar.tech3005@gmail.com

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Sign Up Page



This mockup shows the sign-up page of a website. It features a 'Back to Landing' link in the top left corner. The main content area is a white card with the heading 'AI Image Transformation' and the subtext 'Create your account'. Below this are four input fields: 'First Name', 'Last Name', 'Email address', and 'Password'. The 'Password' field has a toggle icon. Below the 'Confirm Password' field is a 'Sign Up' button. At the bottom of the card, there is a link 'Already have an account? Sign In'. The background of the page is a collage of various comic book characters and scenes.

← Back to Landing

AI Image Transformation

Create your account

First Name Last Name

Email address

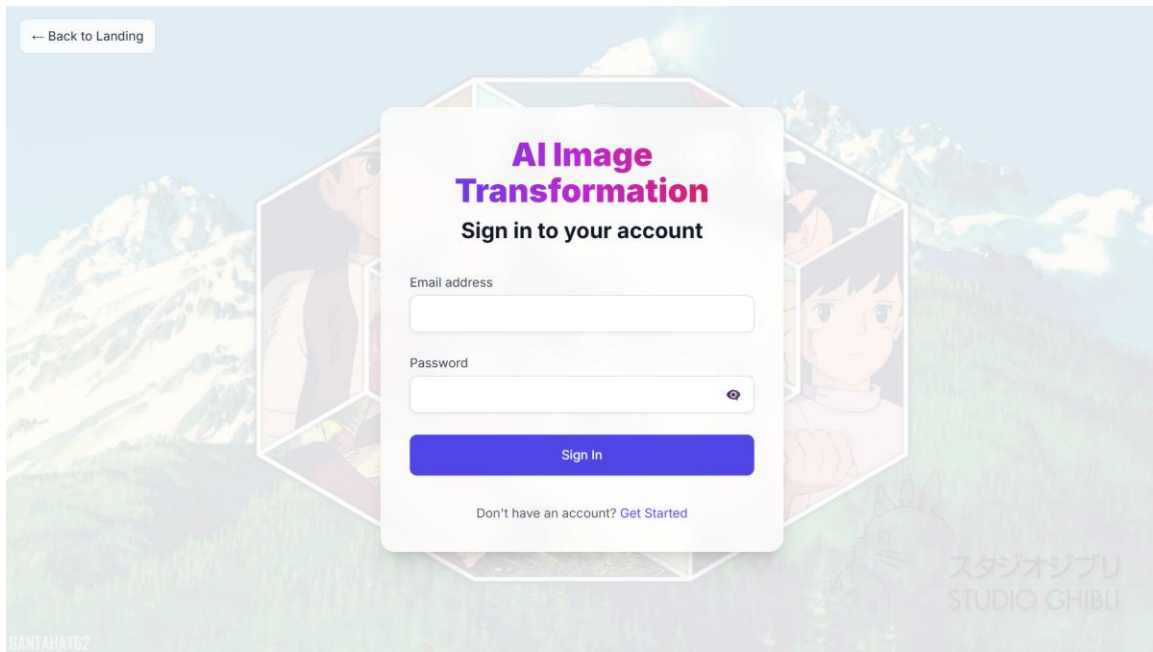
Password

Confirm Password

Sign Up

Already have an account? [Sign In](#)

Sign In Page



← Back to Landing

AI Image Transformation

Sign in to your account

Email address

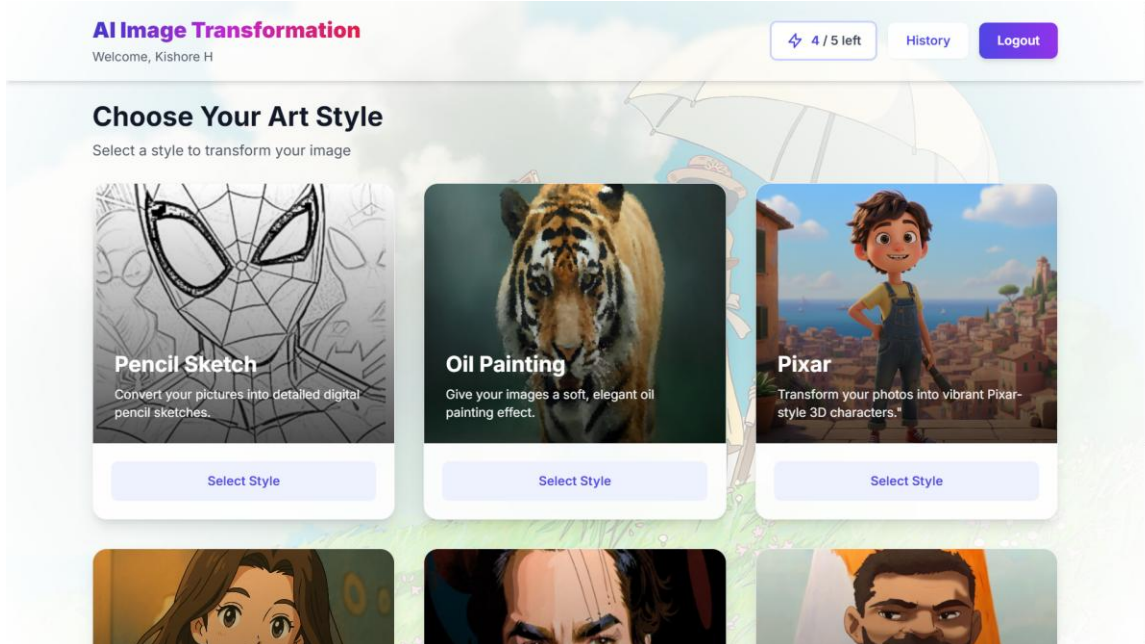
Password

[Sign In](#)

[Don't have an account? Get Started](#)

STUDIO GHIBLI

Dashboard Page



AI Image Transformation
Welcome, Kishore H

4 / 5 left History Logout

Choose Your Art Style

Select a style to transform your image

Pencil Sketch
Convert your pictures into detailed digital pencil sketches.

Select Style

Oil Painting
Give your images a soft, elegant oil painting effect.

Select Style

Pixar
Transform your photos into vibrant Pixar-style 3D characters.*

Select Style

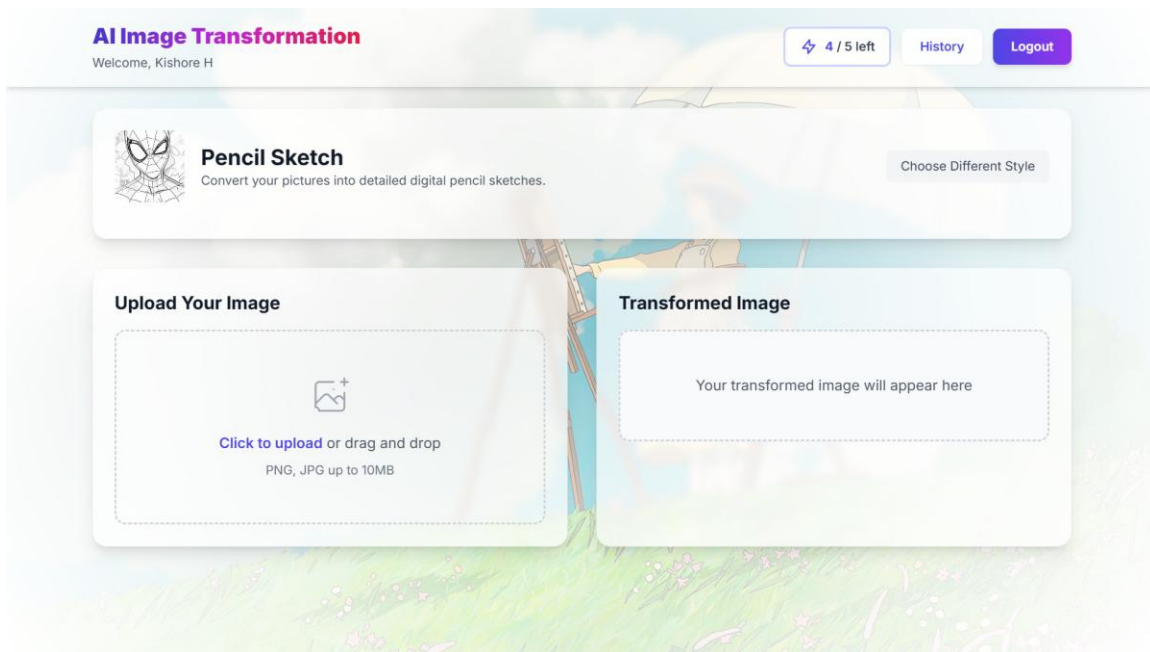
Cartoon
Transform your photos into vibrant cartoon-style characters.

Select Style

3D Model
Transform your photos into vibrant 3D models.

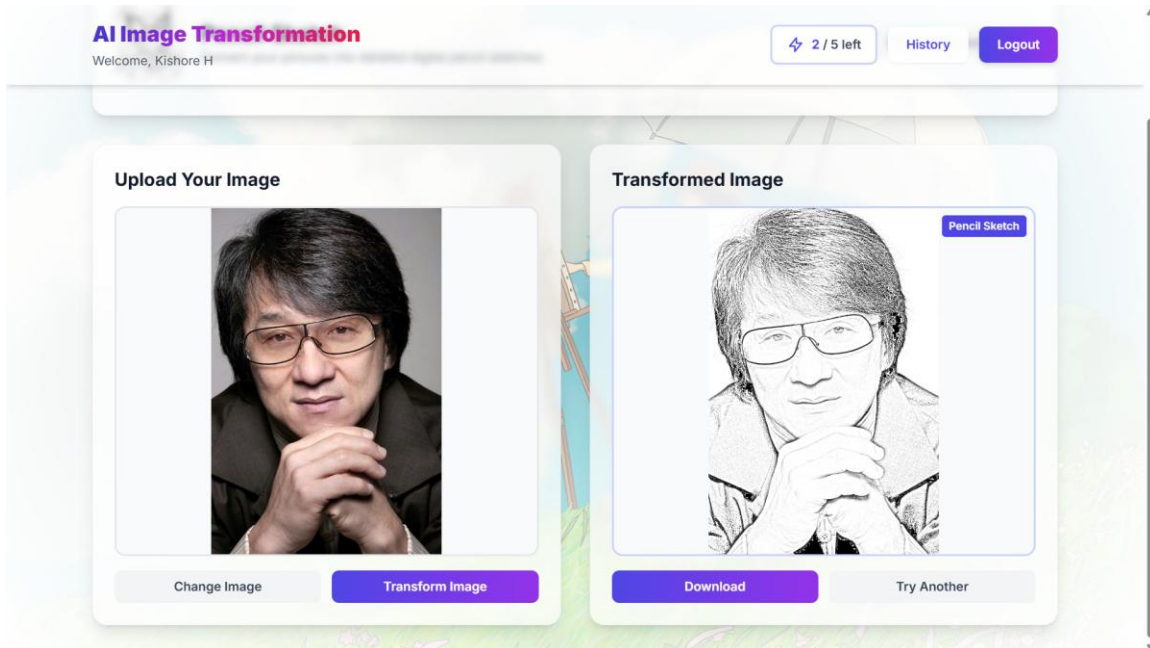
Select Style

Image Transform Page

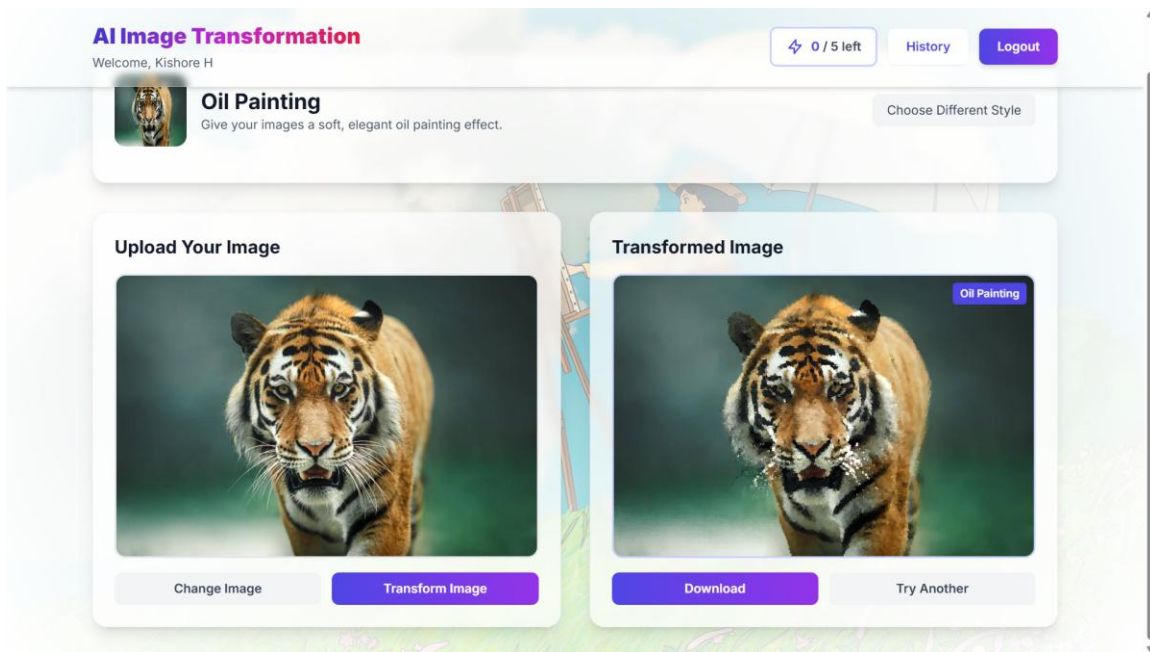


Transformed Results

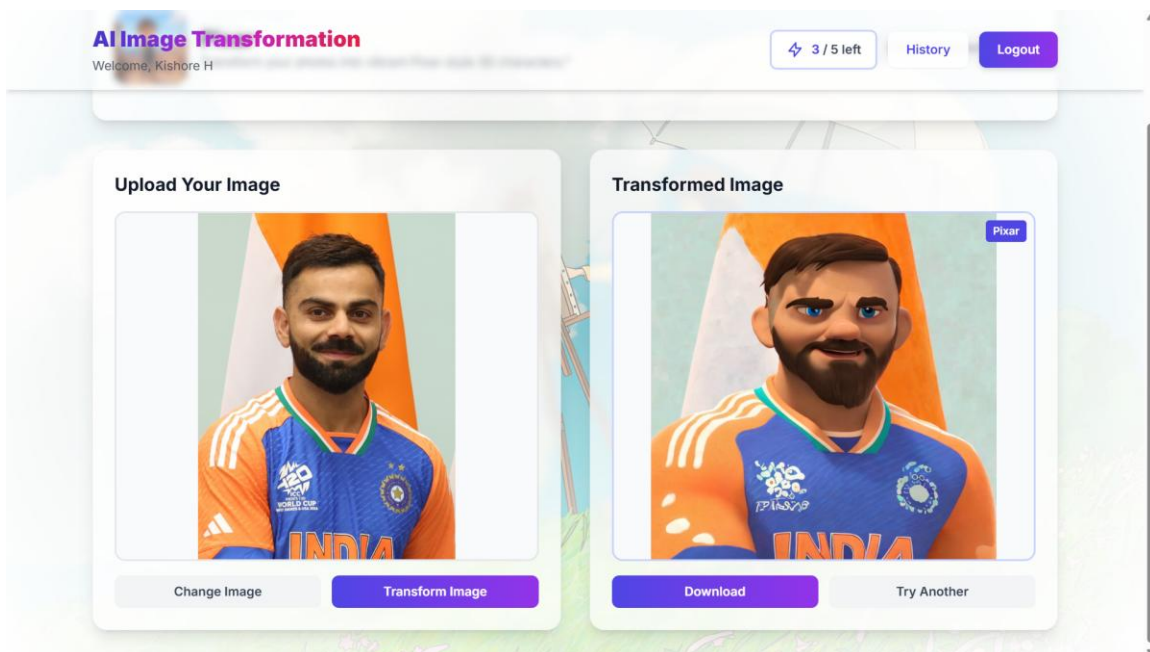
Pencil Sketch



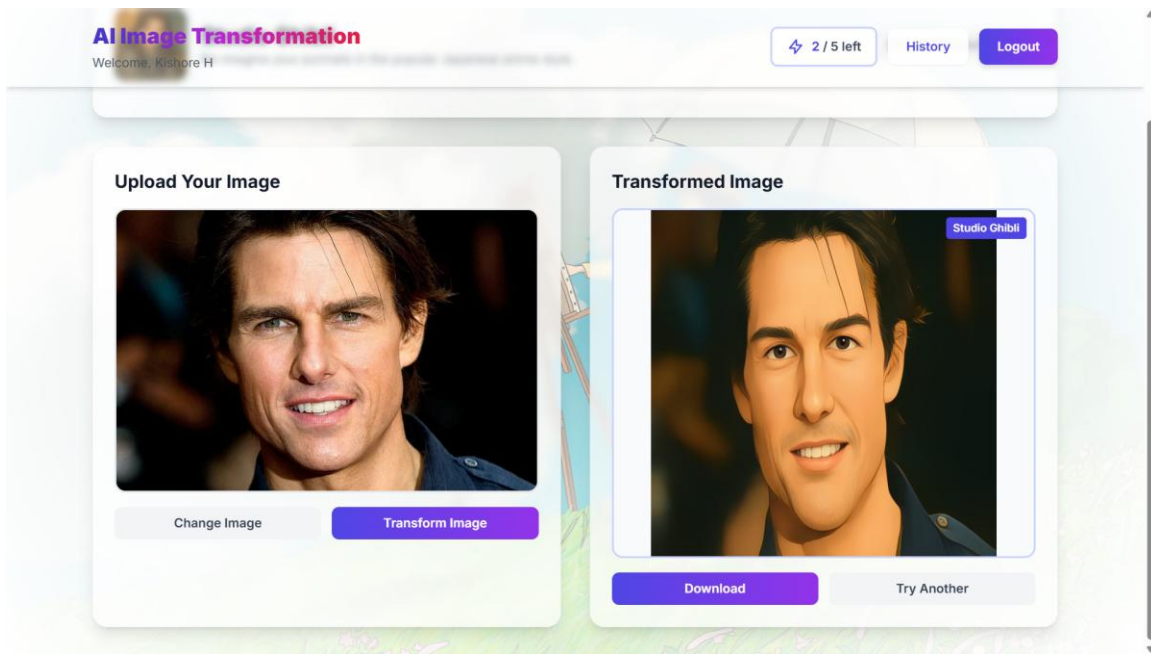
Oil painting



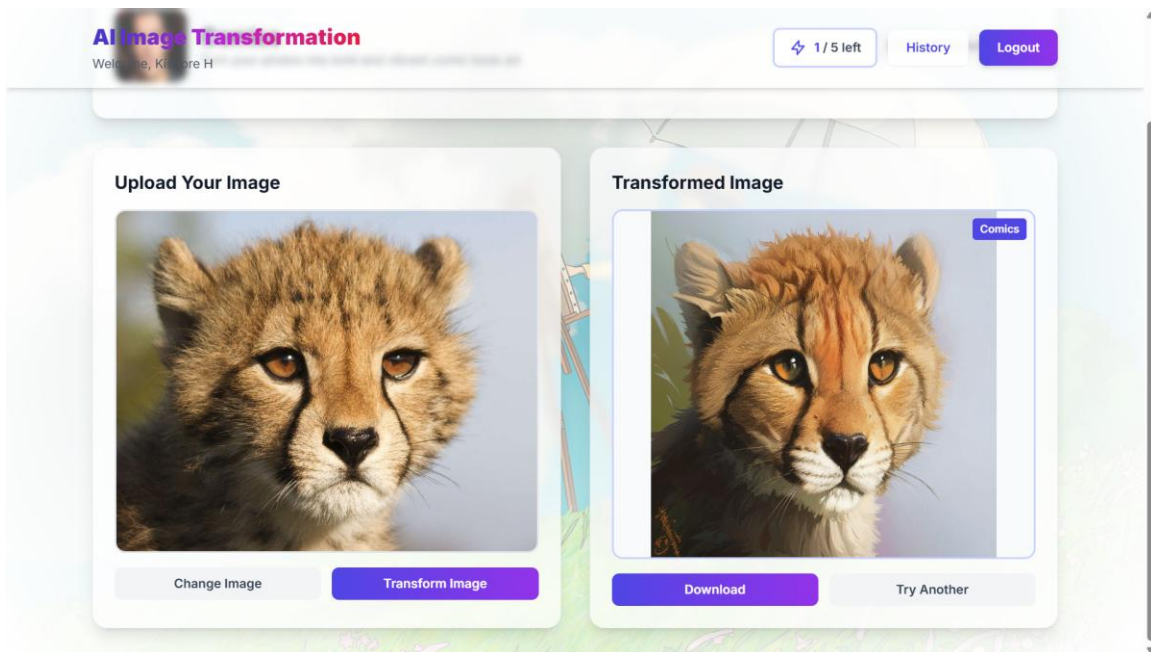
Pixar



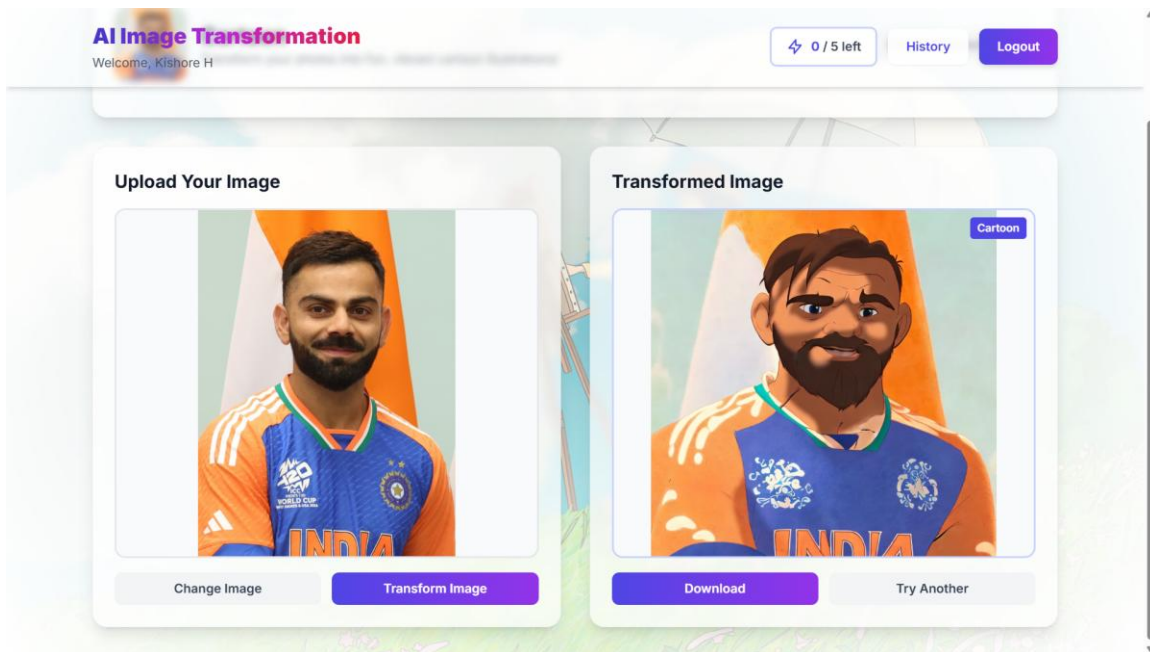
Studio Ghibli



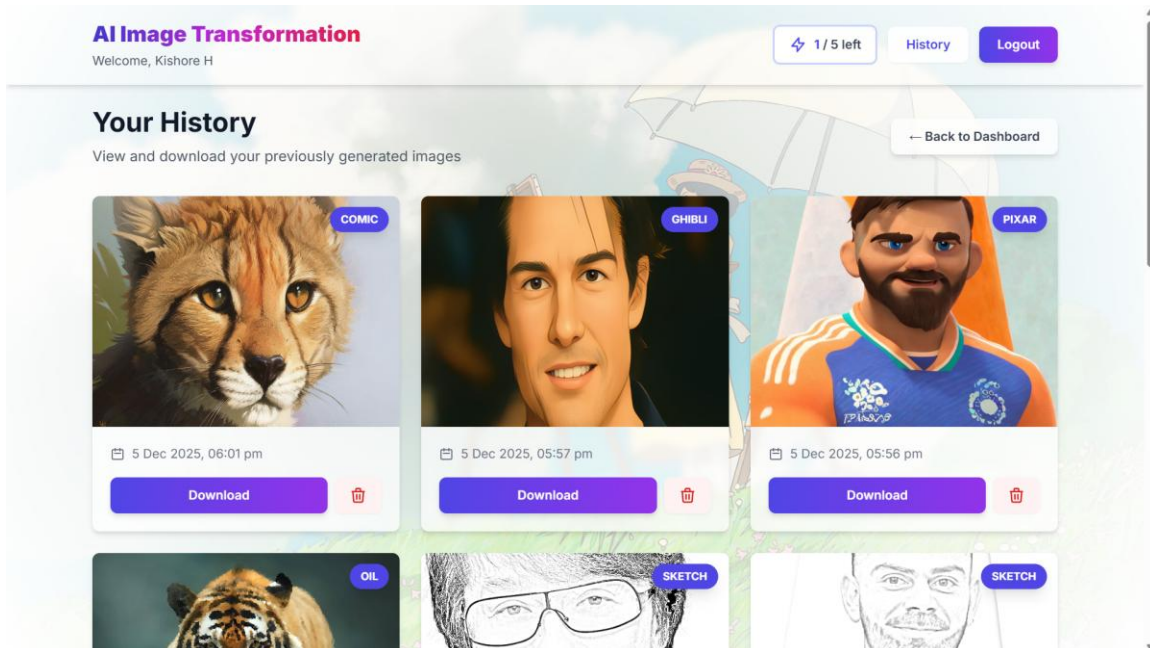
Comics



2D Cartoon

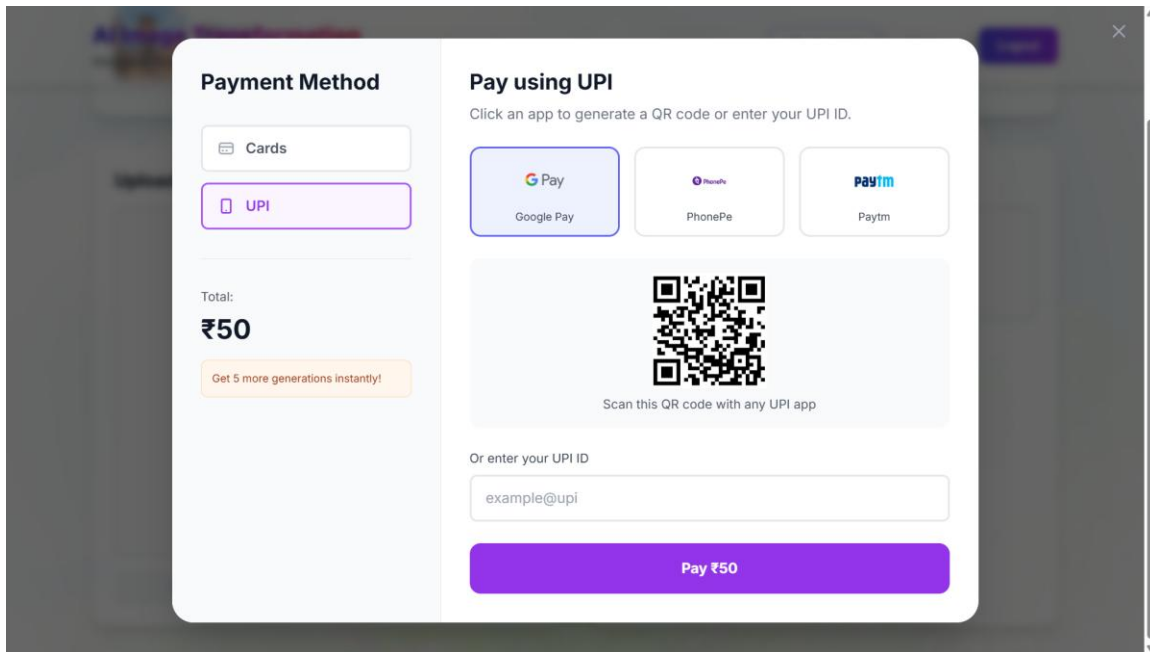


History Page



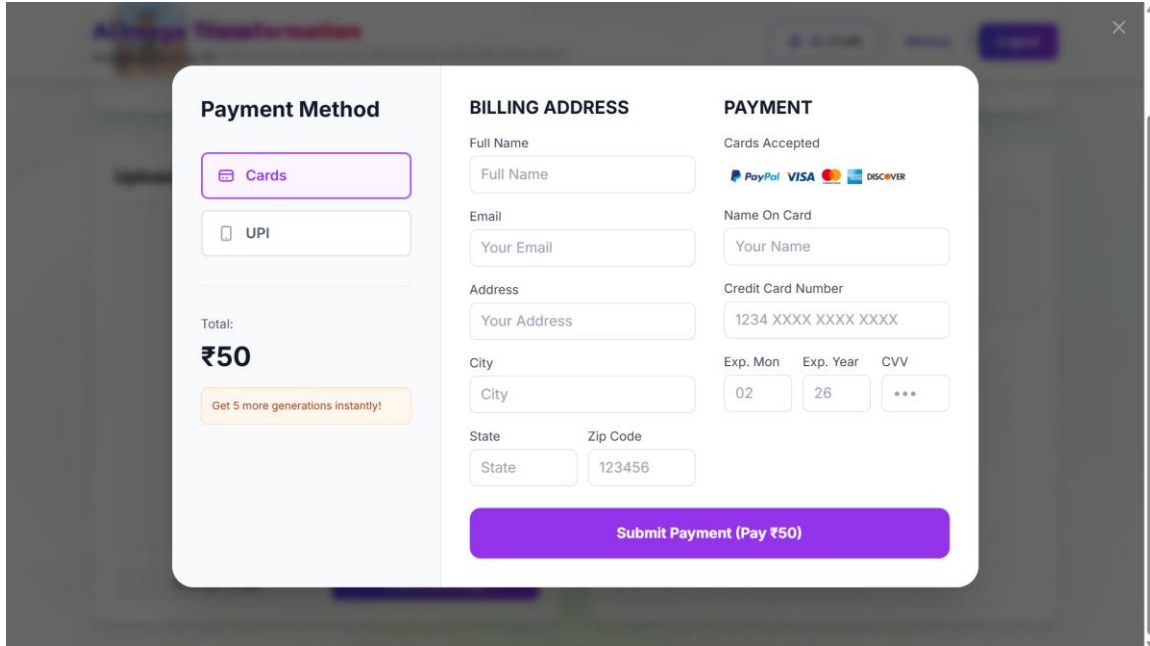
Payment Options

UPI



A screenshot of a 'Payment Method' modal for UPI. On the left, under 'Payment Method', 'Cards' is selected and 'UPI' is highlighted. Below this, the total amount is ₹50, and a promotional banner says 'Get 5 more generations instantly!'. On the right, under 'Pay using UPI', there's a prompt to click an app to generate a QR code or enter a UPI ID. Three app buttons are shown: Google Pay, PhonePe, and Paytm. A large QR code is displayed with the instruction 'Scan this QR code with any UPI app'. Below the QR code is a text input field for 'Or enter your UPI ID' with the placeholder 'example@upi'. At the bottom is a large purple button labeled 'Pay ₹50'.

Card



A screenshot of a 'Payment Method' modal for Card payment. On the left, under 'Payment Method', 'Cards' is highlighted and 'UPI' is selected. Below this, the total amount is ₹50, and a promotional banner says 'Get 5 more generations instantly!'. On the right, under 'BILLING ADDRESS', there are input fields for Full Name, Email, Address, City, State, and Zip Code. Under 'PAYMENT', there's a 'Cards Accepted' section showing logos for PayPal, VISA, Mastercard, and DISCOVER. Below this are input fields for Name On Card, Credit Card Number, Exp. Mon, Exp. Year, and CVV. At the bottom is a large purple button labeled 'Submit Payment (Pay ₹50)'.

7. Challenges Faced

1. **Model Loading Delays** – Heavy Stable Diffusion models caused long initialization times.
Resolution – Implemented lazy loading and cached pipelines to reduce reloading overhead.
2. **GPU Session Limits in Google Colab** – Extended usage caused GPU time-outs, interrupting the execution.
Resolution – Optimized inference steps, reduced unnecessary reloads, and managed runtime usage efficiently to stay within limits.
3. **Difficulty in Selecting Correct Models** – Finding the right models for Pixar, 2D Cartoon, Comics, and Ghibli required testing multiple checkpoints and formats.
Resolution – Performed systematic evaluation and finalized **Nitrosocle/Classic-Anim-Diffusion**, **Dreamshaper8**, **AnimeGAN**, and **OpenCV filters** based on quality and speed.
4. **Ngrok Bandwidth** – ngrok tunnels would disconnect after hitting the bandwidth limits, affecting public access.
Resolution – Restarted tunnels when needed, and optimized data transfer by reducing the payload sizes where possible.

8. Learnings & Skills Acquired

1. Technologies Learned

- I. **Flask** – backend API development.
- II. **SQLite Cloud** – cloud-based data storage.
- III. **Stable Diffusion (Diffusers)** – ML-based image-to-image transformation.
- IV. **AnimeGAN (ONNX Runtime)** – lightweight GPU-optimized image transformation.
- V. **OpenCV, Pillow, SciPy** – image preprocessing & processing.

2. Tools Learned

- I. **Kiro** – AI tool for assisting in coding.
- II. **Google Colab** – Running ML models, GPU-based execution, and hosting.
- III. **Ngrok** – for exposing the backend and frontend services from Google Colab.

3. Soft Skills Acquired

- I. **Multitasking** – Successfully balanced academic examinations alongside daily internship deliverables.
- II. **Time Management** – Completed assigned tasks within deadlines while maintaining consistency.
- III. **Communication Skills** – Effectively communicated progress, clarified requirements, and handled feedback.
- IV. **Team Coordination** – Maintained smooth workflow by coordinating with peers, sharing updates, and supporting each other when needed.

9. Testimonials from team

Throughout the internship, I received **consistent guidance and encouraging feedback** from my mentor and coordinator. They appreciated my ability to **quickly learn new technologies, independently solve issues, and deliver the complete project within the timeline**. Their support and insights played a major role in **refining the overall project quality**.

10. Conclusion

This internship gave me **valuable hands-on experience** in building a complete AI-based image transformation system from scratch. Working across model integration, backend development, frontend development, and cloud database management gave me practical exposure to how full-stack AI applications operate in real environments. The project strengthened my **technical skills in machine learning, diffusion models, and full-stack development, aligning closely with my academic path**. Overall, this experience enhanced both my technical and soft skills and has **better prepared me for future roles** in AI engineering and full-stack application development.

11. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing me with this **valuable internship opportunity**, and an excellent learning platform. I extend my heartfelt thanks to my mentor **K.Munikiran** and coordinator **Nikhil** for their **continuous guidance, support, and constructive feedback** throughout the project. Their insights and encouragement greatly **helped me overcome challenges and successfully complete the internship**.